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Best Practices in Managing Lost Circulation Events in Shuaiba Formation, South Rumaila Field, Iraq in Terms Preventive Measures, Corrective Methods, and Economic Evaluation Analysis

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Abstract

Fluid loss in the Shuaiba formation leads to greater expenses from mud loss, difficulty in well control and zonal isolation, and NPT. This paper will provide a comprehensive statistical work, preventive measures, and corrective actions of lost circulation events in the Shuaiba zone, where this problem is significant. The work will provide an integrated analysis regarding the best ranges of key drilling parameters, which have the greatest chance of avoiding or mitigating lost circulation consequences.

The most significant drilling parameters (e.g. MW, ECD, Yp, RPM, SPM, and ROP) at the time of each event are recorded along with lost circulation remedies attempted, and outcomes of those remedies. Practical information from Rumaila field and range of sources are reviewed and summarized to develop an integrated methodology and flowchart for handling lost circulation events in this zone. This paper will be extended work along with previous comprehensive statistical study and sensitivity analysis models about the Shuaiba formation. In addition, economic evaluation analysis is conducted for partial, severe, and complete losses to obtain the best field procedures.

Proactive approaches are made prior entering the Shuaiba formation to prevent or mitigate the occurrence of the lost circulation. A broad statistical work, primitive mechanisms, typical drilling fluid properties, and recommended operational drilling parameters are estimated to use during drilling through this zone. Moreover, corrective actions are determined for each kind of the mud losses to provide efficient remedies, minimize non-productive time, and reduce cost. The best lost circulation strategy to the Shuaiba formation is concluded and summarized depending on a comprehensive statistical work, the most prevalent industry practices, technical papers, textbooks, and economic analysis evaluation to determine successful remedies for each type of the losses. These treatments are classified by relying on the mud losses classifications to avoid unwanted consequences due to inappropriate actions. In addition, engineered solutions and practical techniques are developed, which will contribute to give clear image and coherent understanding in regard this complicated and costly problem in the Shuaiba zone.

This study provides a unique compilation of information regarding traditional approaches and the latest approaches for lost circulation control. The work provides useful guidelines and possible solutions for all

situations in terms preventive measures, remedial methods, and analytical economic study. In addition, due to the lack of published studies in regard the Shuaiba formation, this work can serve as a vigorous resource for drilling through this zone.

Introduction

Drilling mud accounts for a major expense in drilling oil and gas wells. The drilling mud is circulated through the drill string and drill bit, to remove cuttings from the borehole and to enable drill bit performance. Drilling mud is specifically formulated to develop a thin coating on the borehole wall, referred to as a 'mud cake' which limits fluid losses to the formations already drilled and exposed in the borehole, as drill bit proceeds deeper and deeper.

The concept of lost circulation or lost returns can be defined as "the partial or total loss of circulating fluid from the wellbore to the formation. It is the loss of whole fluid, not simply filtrate, to the formation. Losses can result from either natural or induced causes and can range from a couple of barrels per hour to hundreds of barrels in minutes. Lost circulation is one of the drilling's biggest expenses in terms of rig time and safety. Uncontrolled lost circulation can result in a dangerous pressure control situation and loss of the well" (Baker Hughes, 1999).

Lost circulation is a significant problem in the oil and gas industry. By industry estimates, more than 2 billion USD is spent to combat and mitigate this problem (Arshad et al., 2015). Although it may occur in any formation, some primary contributors to loss circulation are high permeability weakly consolidated formations, fracture calcium carbonate reservoirs and depleted aquifer zones. (Al Menhali et. al, 2015).

Lost circulation may also occur at any point in the drilling operation. If losses occur while drilling a long section of the well, the objective of the treatment will likely be to plug off or limit the losses to allow drilling ahead without casing and cementing. In other situations, the approach may be to limit the losses and cement the well. Given sufficient experience in drilling a particular type of formation, it may be possible to avoid, or significantly minimize lost circulation events by controlling mud properties, drilling rate, or other field parameters. However, this requires a high level of experience and study, which is generally not available. For this reason, the industry relies heavily on using methods of mitigating lost circulation events after they occur.

This study provides basic information on lost circulation, including an introduction to the problem, identifies a range of factors that affect lost circulation, provide proactive techniques, appropriate corrective actions, and economic evaluation analysis to lost circulation in the Shuaiba formation. The study summarizes mud loss and lost circulation information extracted from drilling data from the South Rumaila Field in Iraq. A lost circulation screening criteria are presented for the South Rumaila Field, based on the historical mud loss and lost circulation problems, materials used to mitigate the problems, and potential solutions found by this study.

Shuaiba Formation

The Shuaiba formation occurs at approximately 2900 m and is a limestone with little to no visible porosity. However, the zone is highly susceptible to fracturing and lost circulation, which is more troublesome and even more complicated than lost circulation in the shallower Dammam or the Hartha formations. Sometimes, mud losses in the Shuaiba formation lead to abandonment of the drilling operation due to unsustainable non-productive time (NPT) and drilling cost. Mud losses in Shuaiba formation also cause severe wellbore stability problems. Therefore, it is important that the rig supervisor, mud engineer, log engineer, and geologist meet and develop a remedial action plan, before drilling this formation.

Field methods used to drill the Shuaiba are similar to those noted for the Dammam and Hartha zones, i.e. reduced WOB, RPM, SPM; adjusting mud properties; slow and careful removal or insertion of drill pipe to avoid surging, and breaking gel strength with rotation (South Oil Company, 2008). However, the

level of planning and attention is far higher for this zone because it is the most problematic lost circulation zone among all fields. All rig and field personnel conduct extensive planning prior to drilling the Shuaiba. Increased supplies of lost circulation materials are ensured, and there is precise monitoring of all surface gauges (e.g. bit torque), the shale shaker, desilter, degasser and mud-cleaners. Figure 1 shows the borehole typical of a well drilled through the Shuaiba formation. Both the 13-3/8" and 9-5/8" casing strings have been set. Commonly an 8 1/2" bit is used to drill through the formation.

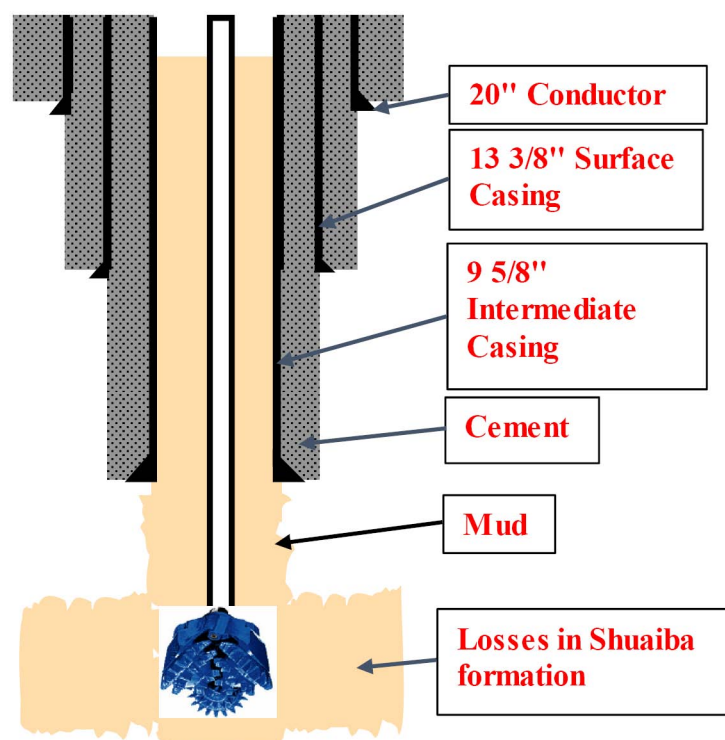


Figure 1—Lost Circulation Mud in Shuaiba Formation

Methodology

Lost circulation events were identified for more than 75 wells discussed in this comprehensive statistical study, according to the formation and depth. Drilling parameters which are known to have the greatest impact and the lowest influence on lost circulation, yet readily adjusted during the drilling operations, were tabulated for analyses (Al-hameedi et al., 2017).

This section presents samples of the wells data, including the number of wells analyzed for the Shuaiba formation. These data have been analyzed to determine ranges for the key drilling parameters and mud properties that have the greatest chance of avoiding or mitigating lost circulation in this formation. All data were collected from various daily drilling report (DDR), final reports, and technical reports. A broad statistical analysis has been conducted to determine which drilling mud properties and operational drilling parameters that have a significant influence on lost circulation (Al-hameedi et al., 2017). This extensive study shows that all mud weight (MW), equivalent circulation density (ECD), and yield point (Yp) have a direct impact on lost circulation whereas SPM, RPM, ROP, WOB and bit nozzles directly or indirectly affect this problem (Al-hameedi et al., 2017). Real data were collected and tabulated to determine the minimum and maximum range of the related parameters to avoid or mitigate lost circulation. Tables 1 and 2 are samples of daily drilling report (DDR) data for loss circulation events gathered for this zone.

Table 1—Well 1 Data Events, Shuaiba Formation

D,(m)	MW, (gm/cc)	YP	SPM	RPM	Nozzles	Type of losses	Type of Treatment	Result
2993 – 3042	1.15	12	85	70	3*12/32	No Loss	No Treatment	Success
3042 – 3088	1.16	13	85	70	3*12/32	Partial Loss	H.V Mud	Success

Table 2—Well 2 Data Events, Shuaiba Formation

D, (m)	MW, (gm/cc)	YP	SPM	RPM	Nozzles	Type of losses	Type of Treatment	Result
3024	1.17	14	80	65	3*12/32	Complete Loss	H.V Mud	Fail
3024 3038	/	/	80	65	3*12/32	Complete Loss	Blind Drilling	Fail
3017	/	/	80	65	No Bit	Complete Loss	Cement Plug	Fail
3017	/	/	80	65	No Bit	Complete Loss	Cement Plug	Fail
3017	/	/	80	65	No Bit	Complete Loss	Cement Plug	Fail
3020 – 3038	/	/	80	65	No Bit	Complete Loss	H.V Mud + Cement Plug	Fail
3021 – 3038	/	/	80	65	No Bit	Complete Loss	H.V Mud + Cement Plug	Fail
3022 – 3038	/	/	80	65	No Bit	Complete Loss	Blend LCM + Cement Plug	Fail

Recommended key drilling parameters have been determined in this paper to prevent or mitigate lost circulation in the Shuaiba formation. This is done based on reviewing data of key drilling parameters. In addition, mud losses treatments events are examined, and statistical analysis is conducted for these remedies. The probability of each treatment is calculated by adding the number of times they were used successfully divided by the total number of attempts. An economic evaluation is performed for the same data based on the cost of each material and the NPT, the rig cost is estimated to be 36000 \$/day. [Table 3](#) shows the prices for lost circulation materials that are used in the economic evaluation ([Halliburton, 2016](#)). Thus, the lost circulation strategy has been developed by depending on statistical work and economic analysis to efficiently remedy in terms stopping mud losses, minimizing non-productive time, and reducing cost. This treatment strategy has been classified by relying on the kind of mud loss. Practical field information from a range of sources was reviewed and summarized to develop an integrated methodology and flowchart for handling lost circulation events in the Shuaiba formation.

Table 3—Cost of Lost Circulation Materials

Material Name	Price for each \$/Ton	Price for each \$/kg
Bentonite	317	0.317
Mica Fine	500	0.5
Mica Medium	700	0.7
Nut Plug	960	0.96
CaCO ₃ Medium	313	0.313
CaCO ₃ Coarse	350	0.35
Super Stop Material	1200	1.2
Blend of LCM	900	0.9
Cement	318	0.318
Diesel Oil	500	0.5

Preventive Measures (Proactive Approaches)

Conventional lost circulation materials (LCMs), including pills, squeezes, pretreatments and drilling techniques often reach their limit in effectiveness and become unsuccessful when drilling deeper hole sections where some formations are depleted, structurally weak, or naturally fractured and faulted (Wang et al., 2005). All those remedies/techniques that are applied prior to entering lost circulation zones to prevent the occurrence of losses can be defined as proactive methods. The main advantage of using these techniques are to increase the chances of avoiding or minimizing lost circulation in the Shuaiba formation. Many methods are used to mitigate mud loss prior to entering the lost circulation zone. Some examples of these methods are waiting method, reduction of pump rate, reduction of mud weight, increase drilling fluid viscosity, and using bit without nozzles (Al-hameedi et al, 2017).

Recommended Key Drilling Parameters to Drill the Shuaiba Formation

Key drilling parameters to drill the Shuaiba formations are summarized in this section using historical data, integrated analysis, and comprehensive statistical study (Al-hameedi et al, 2017).

Mud Weight. This parameter has a pivotal role in lost circulation and it should be monitored carefully while drilling (Al-hameedi et al, 2017). Table 4 shows pore pressure and fracture gradient for the Shuaiba (British Petroleum Company, 2013 and South Oil Company, 2010).

Table 4—Pore and Fracture Gradient for Shuaiba (British Petroleum, 2013)

Formation	Depth, m	PP, (gm/cc)	FP, (gm/cc)	PP, (gm/cc) + Swap Margin	FP, (gm/cc) - Surge Margin
Shuaiba	2990	1.14	1.18	1.15	1.17

Figure 2 shows a plot of volume loss versus mud weight for more than 75 wells drilled through the Shuaiba formation. The data show a noticeable increase in losses when the mud weight exceeds 1.16 gm/cc. From this plot, the optimal mud weight to drill the Shuaiba formation is 1.15 gm/cc to 1.16 gm/cc.

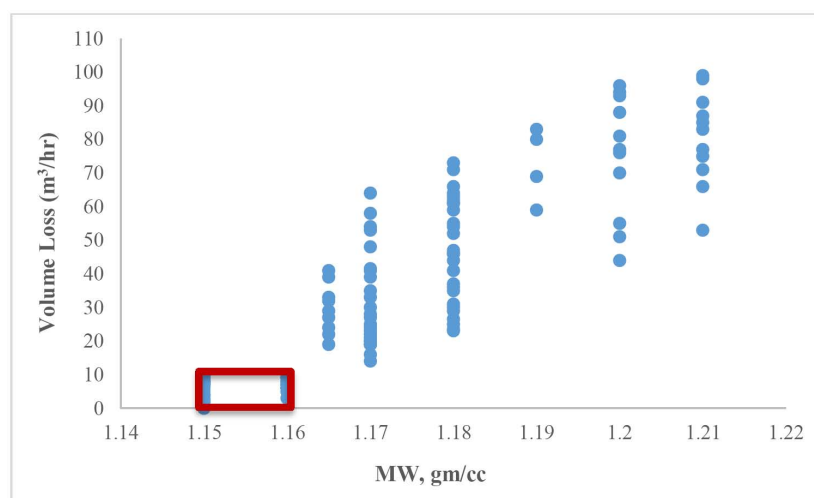


Figure 2—Mud Weight versus Volume Loss (Shuaiba Zone, more than 75 wells)

Equivalent Circulation Density (ECD). This property should be monitored during drilling operation since it has a big role on lost circulation (Al-hameedi et al, 2017).

Figure 3 shows a plot of volume loss versus ECD for more than 75 wells drilled through the Shuaiba formation. The data show a noticeable increase in losses when the ECD exceeds 1.18 gm/cc. From this plot, the proper equivalent circulation density to drill the Shuaiba formation is 1.16 gm/cc to 1.18 gm/cc.

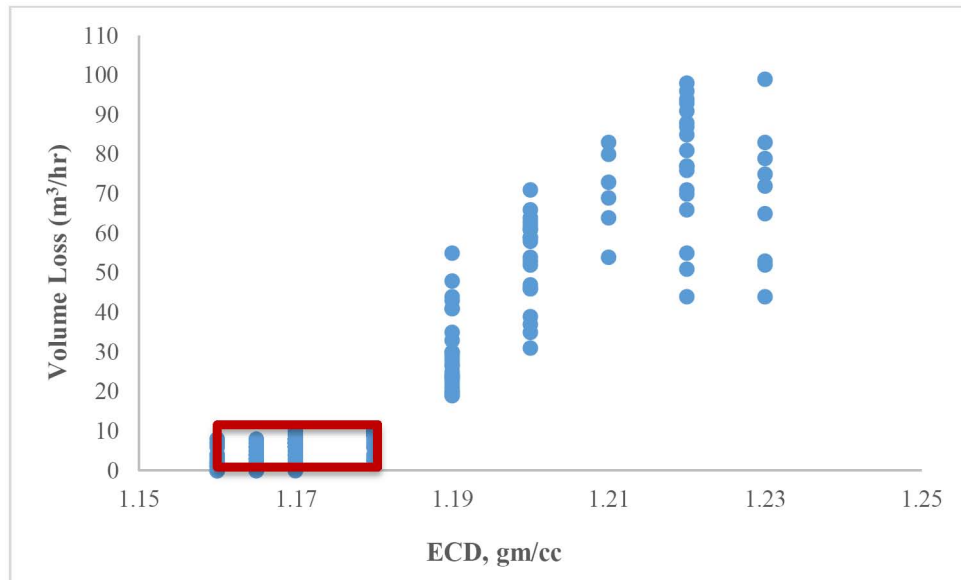


Figure 3—Equivalent Circulation Density (ECD) versus Volume Loss (Shuaiba Zone, more than 75 wells)

Yield Point (Yp). Efficient hole cleaning is largely relying on yield point. In other words, this property is responsible for suspending and lifting cutting to the surface (Al-hameedi et al, 2017). From figure 4, the proper yield point (Yp) that should be used to drill through the Shuaiba zone is from 12 lbf/ft² to 13 lbf/ft². These values will provide efficient hole cleaning, decrease ECD, and minimize losses pressure in the annulus.

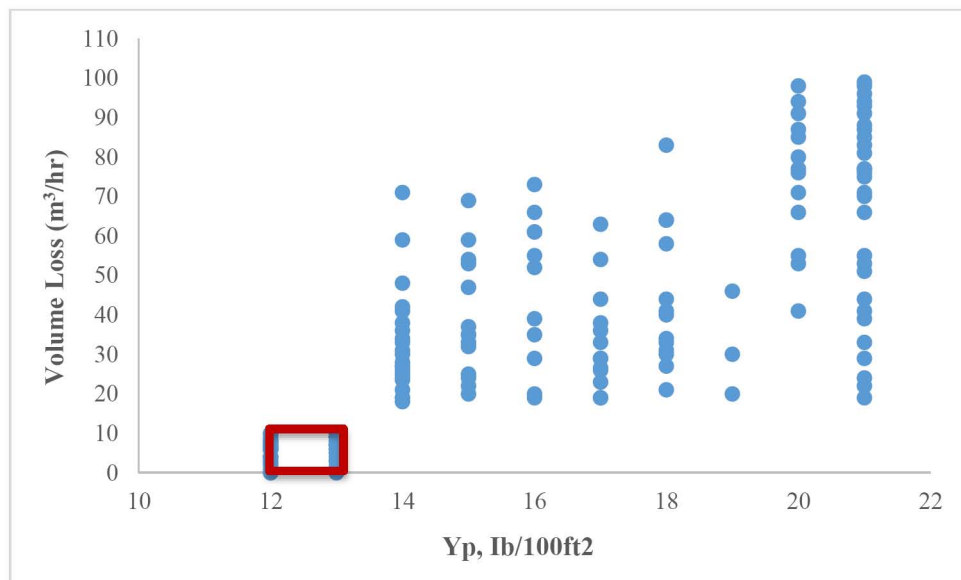


Figure 4—Yield Point (Yp) versus Volume Loss (Shuaiba Zone, more than 75 Wells)

Plastic Viscosity (PV). This parameter is related to effective drilling density. It considers the second component of the drilling fluid viscosity (Al-hameedi et al, 2017). Figure 5 shows a plot of volume loss versus plastic viscosity (PV) for more than 75 wells drilled through the Shuaiba formation. From this plot, the optimal plastic viscosity (PV) to drill the Shuaiba formation is 12 cp to 15 cp.

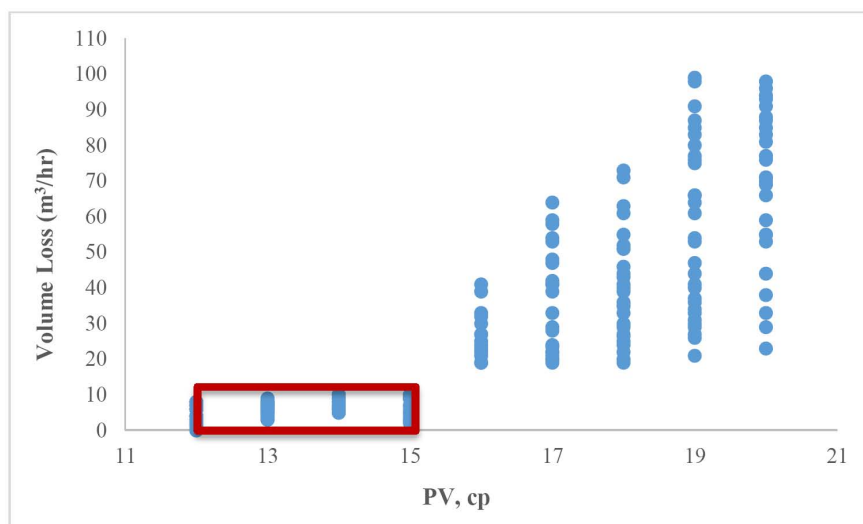


Figure 5—Plastic Viscosity versus Volume Loss (Shuaiba Zone, more than 75 Wells)

Weight on Bit (WOB). This parameter has a significant impact on rate of penetration. By increasing weight on bit, rate of penetration (ROP) will be maximized; therefore, effective mud weight will be increased. Hence, weight of bit has directly or indirectly influence on mud loss. Figure 6 shows a plot of volume loss versus WOB for more than 75 wells drilled through the Shuaiba formation. From this plot, the proper WOB to drill the Shuaiba formation is 10 Ton to 14 Ton.

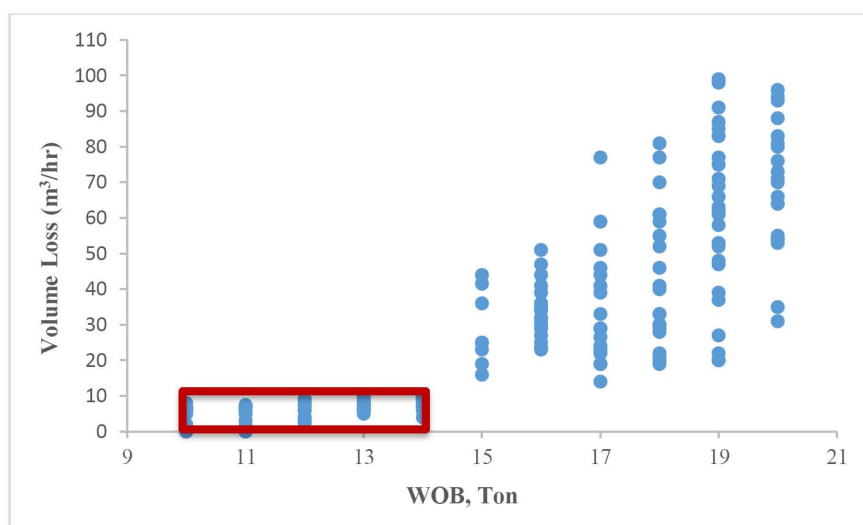


Figure 6—Weight on bit (WOB) versus Volume Loss (Shuaiba Zone, more than 75 Wells)

Strokes per Minute (SPM) & Flow Rate (Q). Both parameters are related to mud pump pressure. They are responsible for drilling mud cycle from mud system to wellbore by using mud pumps. (Al-hameedi et al, 2017).

Figures 7 and 8 show plots of volume loss versus strokes per minute (SPM) and flow rate (Q) respectively for more than 75 wells drilled through the Shuaiba formation. The data show a noticeable increase in losses when the SPM and Q exceed 90 and 1584 (L/STK) respectively. From these figures, the proper strokes per minute and flow rate that should be used to drill the Shuaiba zone are from 80 SPM to 90 SPM and from 1408 L/STK to 1584 L/STK respectively.

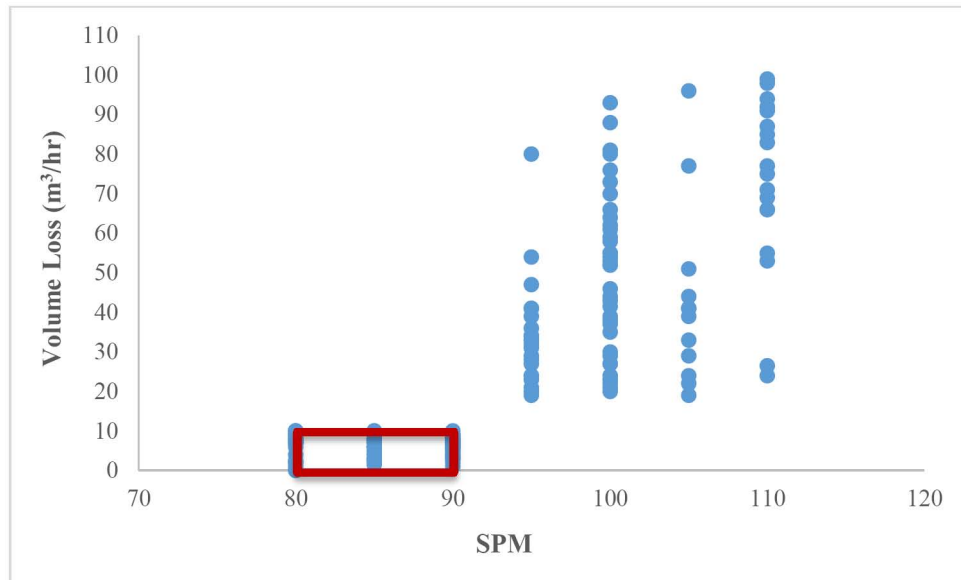


Figure 7—Stroke per Minute (SPM) versus Volume Loss (Shuaiba Zone, more than 75 Wells)

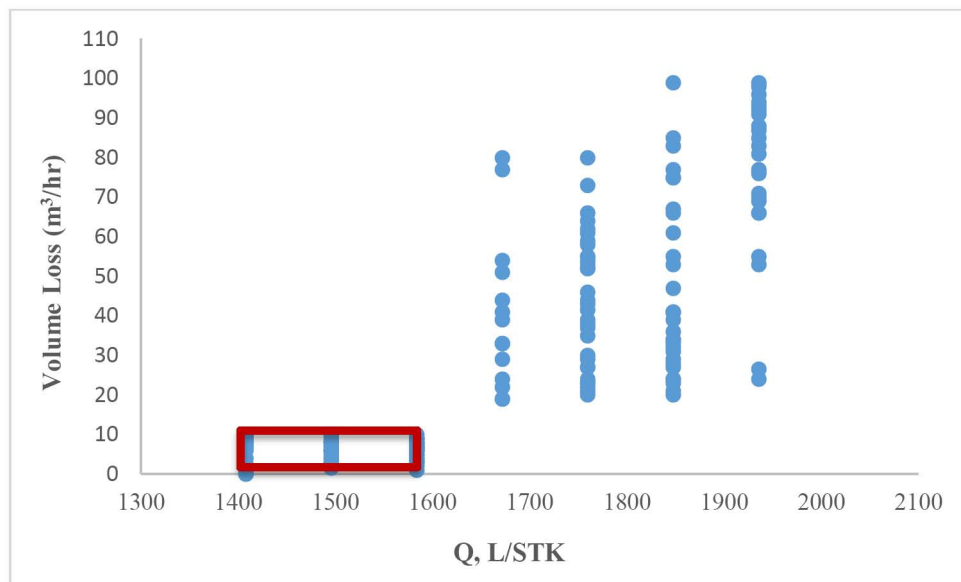


Figure 8—Flow rate (Q) versus Volume Loss (Shuaiba Zone, more than 75 Wells)

Revolutions per Minute (RPM). This property is related to rotate drill string, bit, and penetration rate. Figure 9 shows the relationship between revolutions per minute (RPM) and volume loss for the Shuaiba zone.

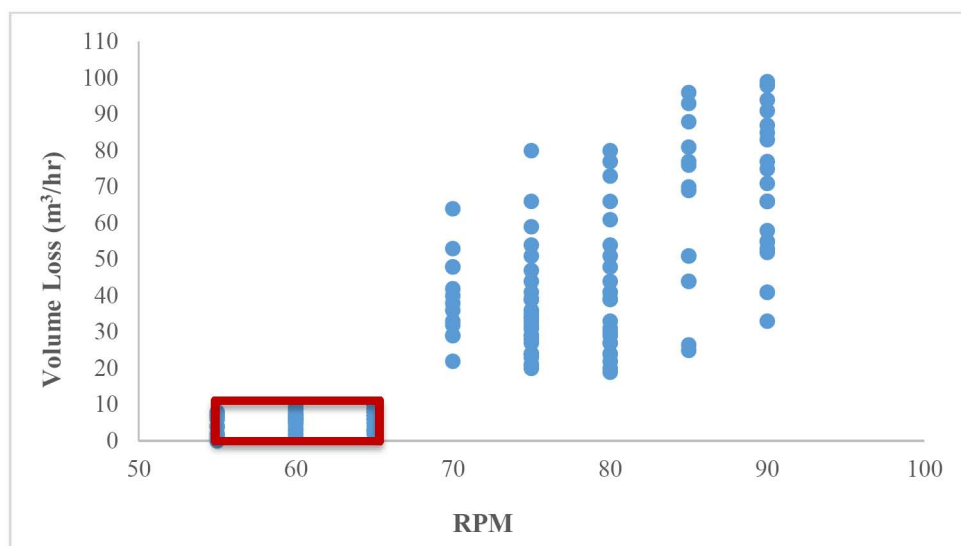


Figure 9—Revolutions per Minute (RPM) versus Volume Loss (Shuaiba Zone, more than 75 wells)

From Figure 9, the appropriate RPM that is advisable to drill the Shuaiba zone is from 55 RPM to 65 RPM. This range will provide a good penetration rate, decrease cutting amounts, and minimize friction pressure into the annulus.

Rate of Penetration (ROP). Excessive cutting and high rate penetration will lead to increase downhole pressure (Al-hameedi et al, 2017). Figure 10 shows a plot of volume loss versus ROP for more than 75 wells drilled through the Shuaiba formation. The data show a noticeable increase in losses when the ROP exceeds 4 m/hr. From Figure 10, the proper ROP that should be used to drill the Shuaiba zone is from 2 to 4 m³/hr.

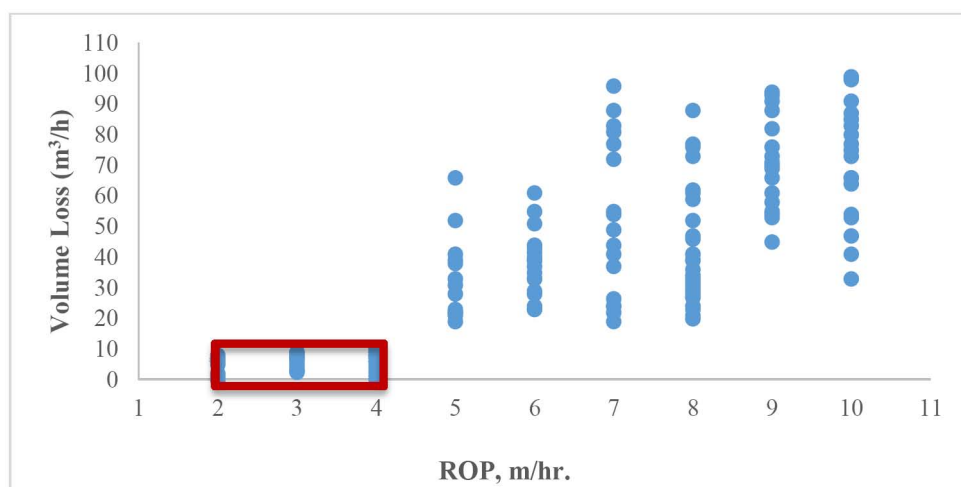


Figure 10—Rate of Penetration (ROP) versus Volume Loss (Shuaiba Zone, more than 75 wells)

Bit Without Nozzles (WON). It is advisable to use bit without nozzles during drilling the Shuaiba formation for several reasons like to reduce jet velocity on the formation, minimize non-productive time (NPT), to use any type of lost circulation mud (LCMs), and to avoid nozzles plugging (Al-hameedi et al, 2017). Figure 11 shows the relationship between total flow area of bit nozzles and volume loss for more than 75 wells drilled through the Shuaiba formation. From this figure, it is easy to see that the use of bit without nozzles will have the lowest mud losses. Therefore, it is recommended to use bit without nozzles when drilling the Shuaiba formation.

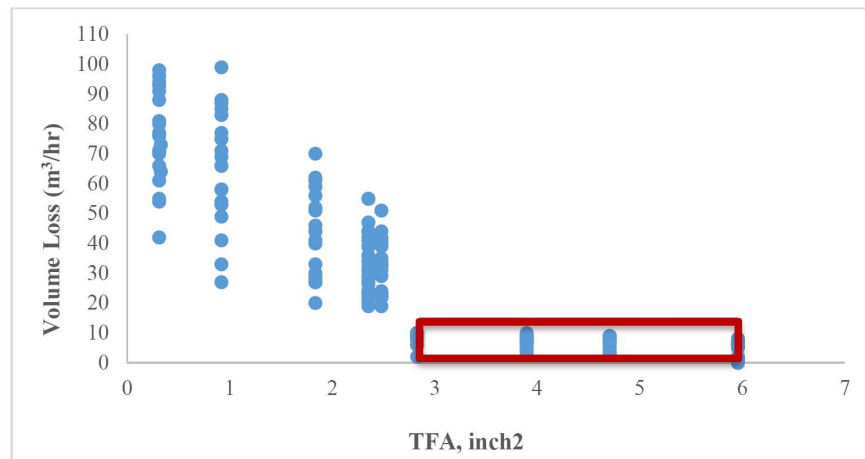


Figure 11—Total Flow Area of Bit Nozzles versus Volume Loss (Shuaiba Zone, more than 75 wells)

Table 5 shows recommended parameters to drill the Shuaiba formation. In some cases, under the same recommended parameters, Shuaiba zone will suffer from severe mud loss problem or even complete losses. The major reason these losses occur is due to using a high range of key drilling parameters before using the recommended parameters. If the recommended key drilling parameters are not utilized initially and the lost circulation problem occurs, the key drilling parameters can't be used afterwards and the only option left is to use treatments to mitigate or stop losses.

Table 5—Recommended Key Drilling Parameters for Dammam Formation

Property	Minimum Value	Maximum Value
Mud Weight (MW, gm/cc),	1.15	1.16
Equivalent Circulation Density (ECD), (gm/cc)	1.16	1.18
Yield Point (Yp), (lbf/100ft ²)	12	13
Plastic Viscosity (PV), cp	12	15
Weight of Bit (WOB), Ton	10	14
Strokes per Minute (SPM)	80	90
Flow Rate (Q), L/STK	1408	1584
Revolutions per Minute (RPM)	55	65
Rate of Penetration (ROP, m/hr.)	2	4
Bit Nozzles	Without Nozzles	Without Nozzles

Corrective Methods (Remedial Treatments)

If the key drilling parameters from the proactive approach are not utilized and the mud losses occur, then the only option left is to use corrective methods and treatments to treat mud losses. On this section, several remedies that have been used to treat mud losses for more than 75 wells in the Shuaiba formation are presented. Each type of remedy requires a specific procedure to apply it on the field. [Al-hameedi et al \(2017\)](#) presented detailed procedures for partial, severe, and complete losses treatments that were used in the South Rumaila field. Selecting appropriate treatment by depending on the type of the lost circulation, which will reflect positively on the drilling operations in terms combating the problem, saving time, and

reducing expenses. In this section, remedies will be classified depending on the type of the losses ([South Oil Company, 2010](#)).

Partial Losses Remedies

A comprehensive statistical study has been conducted to determine the optimal treatments to stop this type of loss. [Figure 12](#) shows the probability of success and failure for the recommended actions which should be used to treat partial losses.

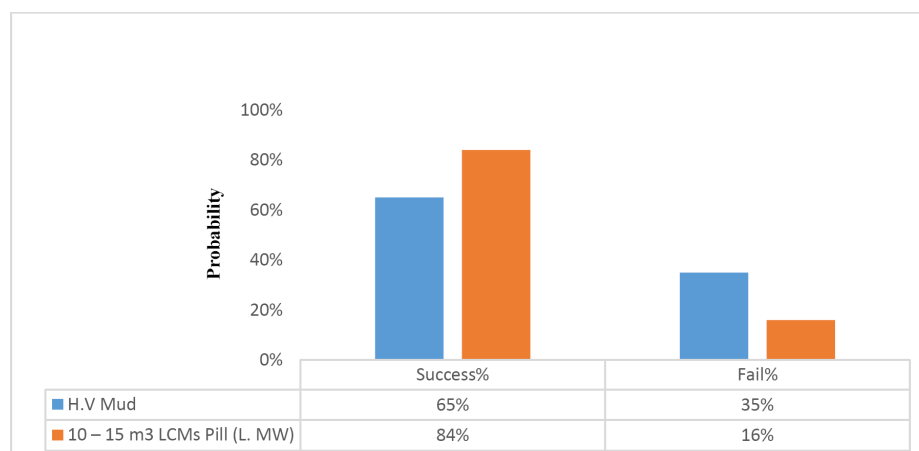


Figure 12—Recommended Remedies for Partial Losses (Shuaiba Zone, more than 75 wells)

Severe Losses Remedies

A comprehensive statistical study has been conducted to determine the optimal treatments to stop this type of losses. [Figure 13](#) shows the probability of success and failure for the treatments that were used to treat severe losses in the Shuaiba formation.

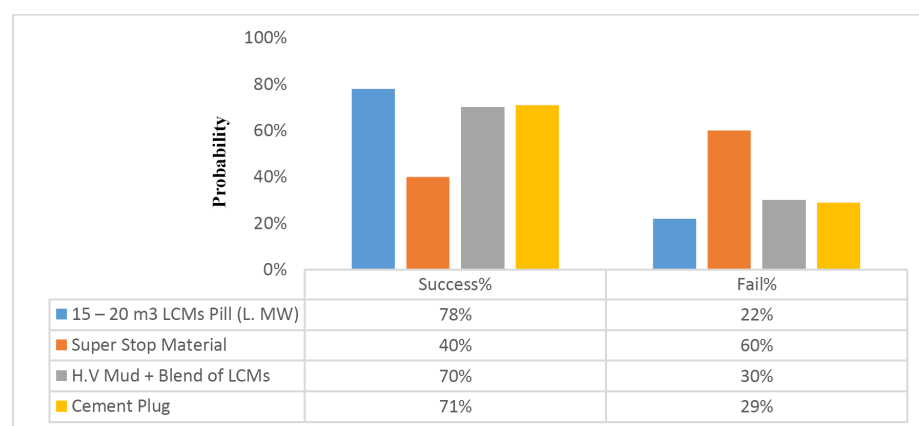


Figure 13—Recommended Remedies for Severe Losses (Shuaiba Zone, more than 75 wells)

Complete Losses Remedies

[Figure 14](#) shows the probability of success and failure for treatments that were used to treat complete losses in the Shuaiba formation.

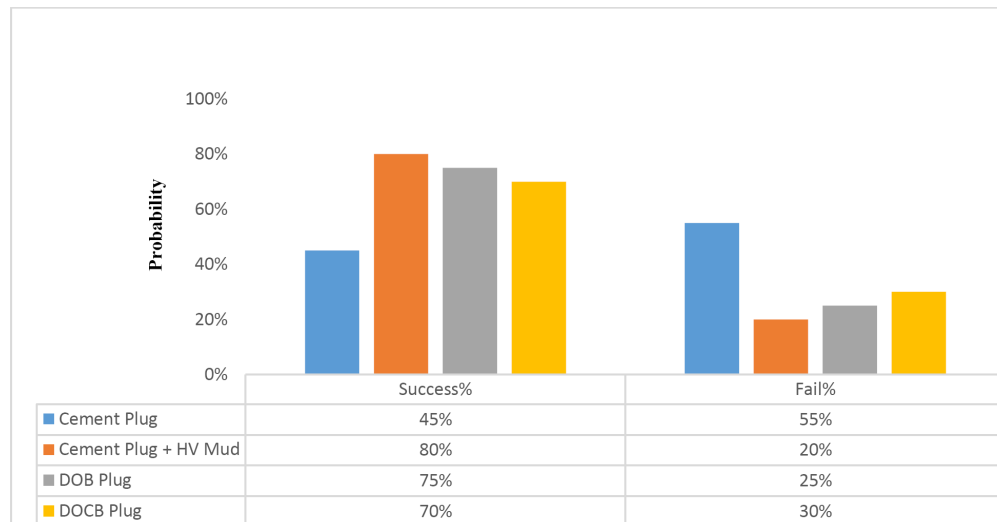


Figure 14—Recommended Remedies for Complete Losses (Shuaiba Zone, more than 75 wells)

Economic Evaluation

The economic evaluation is conducted for partial, severe, and complete losses. Table 6 shows the results of the economic evaluation for the best partial losses treatments with their probabilities, pill of LCM treatment has a higher probability of success than the H.V Mud Patch treatment.

Table 6—Partial Losses Economic Calculations and Probabilities

Treatment Name	Required Addition, kg/m ³	Cost, \$/m ³	Waiting Period, (hrs)	NPT Cost, \$/1hr	Total Cost, (\$)	Success%	Fail%
H.V Mud Patch	100	31.7	2.5	1500	3781.7	65	35
Pill of LCM	Mica Fine (15), Mica Medium (15), Nut Plug (15), CaCO ₃ Medium (15), CaCO ₃ Coarse (15)	42.345	3.5	1500	5292.345	84	16

Table 7 shows the economic calculations for the best severe losses treatments, pill of LCM treatment has the highest probability of success for the severe losses.

Table 7—Severe Losses Economic Calculations and Probabilities

Treatment Name	Required Addition, kg/m ³	Cost, \$/m ³	Waiting Period, (hrs)	NPT Cost, \$/1hr	Total Cost, (\$)	Success%	Fail%
Pill of LCM	Mica Fine (30), Mica Medium (30), Nut Plug (30), CaCO ₃ Medium (30), CaCO ₃ Coarse (30)	84.69	3.5	1500	5334.69	78	22
Super Stop Material	125	150	4.5	1500	6900	40	60
H.V Mud + Blend of LCM	Bentonite (100), Blend LCM (45)	72.2	5	1500	7572.2	70	30
Cement Plug	1176	373.97	18	1500	27374	71	29

Economic evaluation and probabilities for the best complete losses treatments are shown in Table 8, H.V Mud + Cement Plug treatment has the highest probability of success.

Table 8—Complete Losses Economic Calculations and Probabilities

Treatment Name	Required Addition, kg/m ³		Cost, \$/m ³	Waiting Period, (hrs)	NPT Cost, \$/1hr	Total Cost, (\$)	Success%	Fail%
Cement Plug	1176		374	18	1500	27374	45	55
H.V Mud +Cement Plug	Bentonite (100), Cement (1029)		405.67	20	1500	30405.67	80	20
DOB Plug	Formula for 1 m ³		603.6	10	1500	15603.6	75	25
	Oil base	0.70 m ³						
	Bentonite	800 kg						
DOBC Plug	Formula for 1 m ³		645.75	12	1500	18645.75	70	30
	Oil base	0.72 m ³						
	Bentonite	450 kg						
	Cement	450 kg						

Recommended Lost Circulation Strategy to the Shuaiba Formation

This section summarizes the required treatments for each type of mud losses. Figure 15 presents a flow chart that can be used to treat mud losses in the Shuaiba formation. It was developed by reviewing the historical data of more than 75 wells drilled in the South Rumaila Field, statistical analysis, and economic evaluation. The flow chart considers the use of the highest probability of success treatment at the beginning even if that treatment is more expensive than other for the following reasons:

- To maximize the guarantee of the treatment success.
- To avoid or reduce repetition of the treatments that use to stop lost circulation.
- To minimize Non-Productive time by using appropriate actions.
- To acquire more effectively cost. In different words, using corrective measures that are associated with high success percentage are more economic than applying remedial actions that have low success percentage.

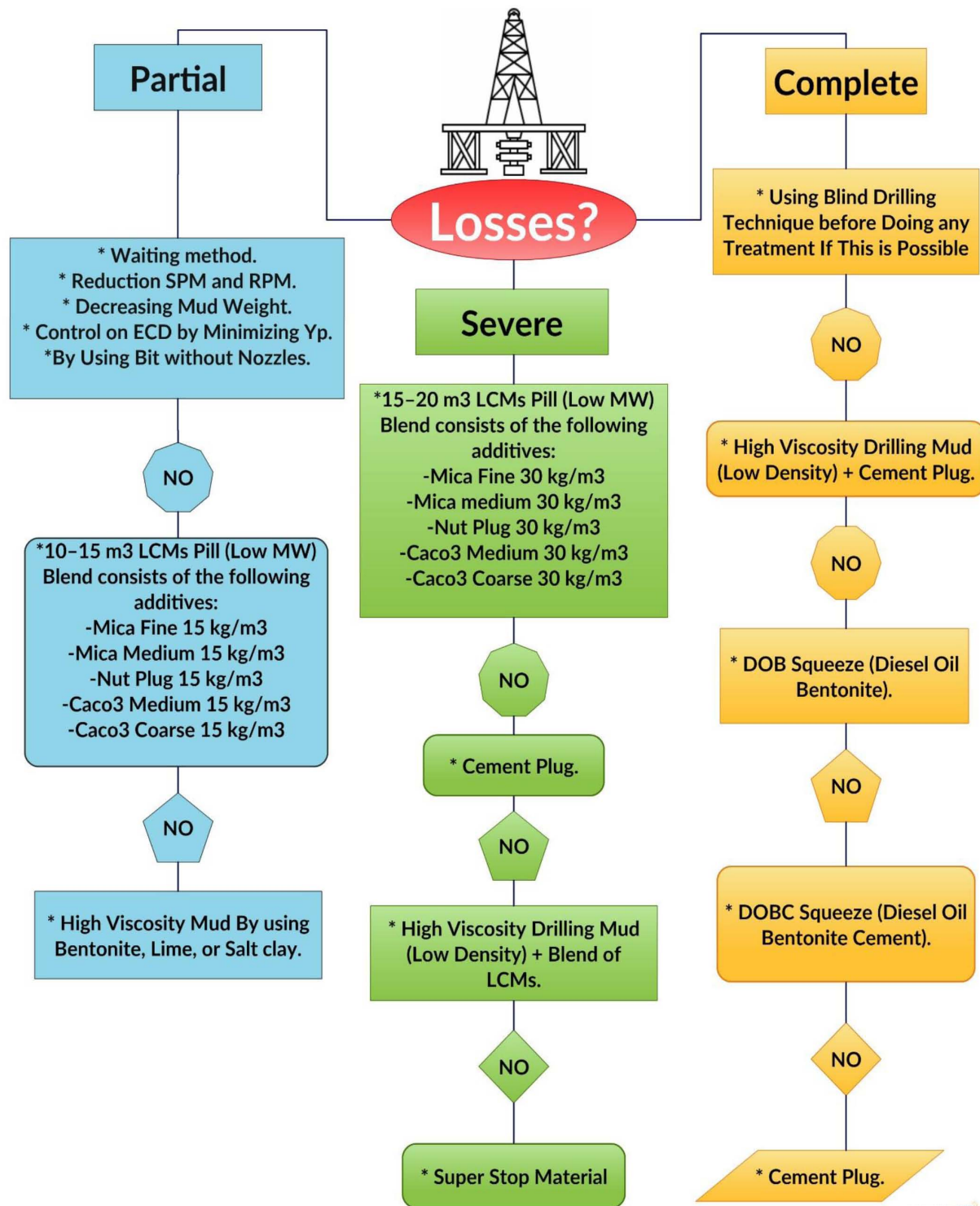


Figure 15—Treatment Strategy to Shuaiba Formation

Figure 15 provides guidance for drilling through the Shuaiba formation and handling loss circulation through this zone.

Conclusion

This paper has provided a detailed study of lost circulation, including a brief review of fundamentals of lost circulation, analyzing real field data, discussion of methods of mitigating losses, and an introduction to newer methods of loss control used in industry. Based on this study, the following conclusion are made:

- Key drilling parameters that should be used to drill the Shuaiba formation are identified and summarized in a table. Lost circulation can be avoided or mitigated when using these parameters.
- One challenge in drilling wells in the Rumaila field is the inconsistency of approaches to the lost circulation problem. Hence, a formalized methodology for responding to losses in Shuaiba zone is developed and provided as means of assisting drilling personnel to work through the lost circulation problem in a systematic way.
- Treatments for partial, severe, and complete losses for the Shuaiba formation are summarized in a flow chart. This flow chart should be used to treat the mud losses in the Shuaiba formation depending on the type of mud losses.
- The highest probability of success treatment should be used to treat the mud losses even if it is not the cheapest to avoid the repetition of treatments which reduces the NPT. Using a low-probability of success treatment may not be effective and the usage of multiple treatments may be required, even if it is cheaper than other treatment but the NPT will be higher which increases the cost.
- The first treatment that should be used to treat complete losses in the Shuaiba formation is blind drilling. This method is dangerous and should be applied carefully in the field.
- The first treatment that should be used to treat partial losses is the waiting method. If it fails, then use the recommended treatments in the flow chart.
- The first treatment that should be used to treat severe losses is high viscosity mud and blend of LCM. If it fails, then use the recommended treatments in the flow chart.

Acknowledgments

The authors would like to thank South Oil Company from Iraq and British Petroleum Company for providing us various real field data.

Nomenclature

APL	Annular Pressure Los
bbl/hr	barrels per hour
BPC	Basra Petroleum Company
cp	centipoise
D	Depth
DDR	Daily Drilling Report
DOH	Diameter of Open Hole
ECD	Equivalent Circulation Density
FCL	Ferro Chrome Lignosulfonate
FP	Fracture Pressure
Ft/min	foot per minute
FWB	Fresh Water Bentonite
gm/cc	gram per cubed centimeter
HP	Hydrostatic Pressure
H. V	High Viscosity
Ib/bbl	pounds per barrel
Ib/ft ³	pounds per cubed feet

in	Inch
Kg/m ³	Kilogram per cubed meter
LCMs	Lost Circulation Materials
L/min	Litter per minute
m	meter
m ³ /hr	cubed meter per hour
MW	Mud Weight
NPT	Non-productive Time
O.E.D.P	Open End Drill Pressure
ppg	pounds per gallon
PP	Pore Pressure
Q	Flow Rate
ROP	Rate of Penetration
RPM	Revolutions per Minute
SPM	Stroke per Minute
TFA	Total Flow Area
WOB	Weight of Bit
WOC	Waiting of Cement
WON	Without Nozzles
Yp	Yield Point Viscosity
\$	Dollar

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